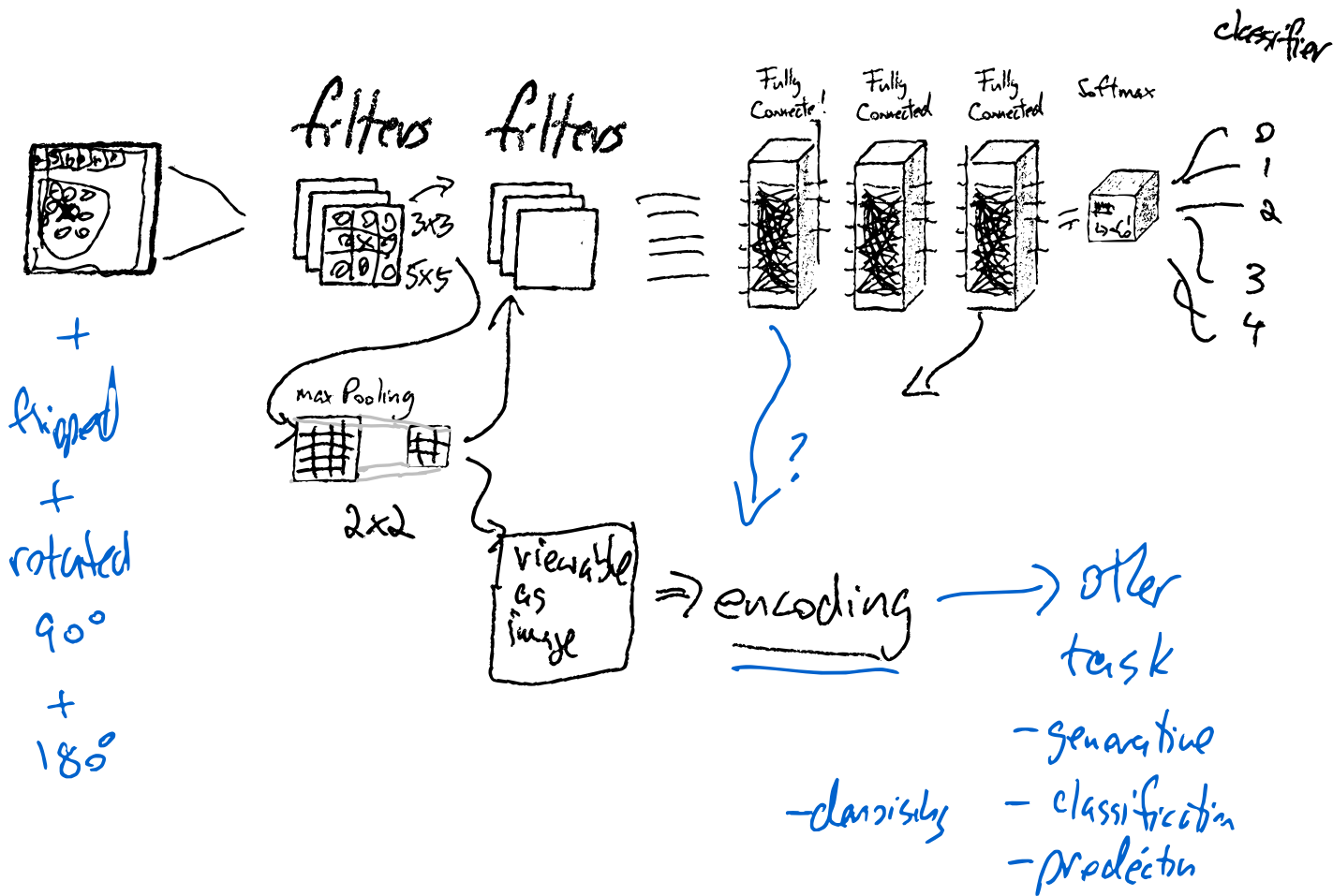




Hierarchical Filter Learning

Label y : $\begin{matrix} 0 \\ \text{(Indoors)} \\ \uparrow \uparrow \uparrow \end{matrix}$ $\begin{matrix} 1 \\ \text{(Outdoors)} \\ \uparrow \uparrow \uparrow \end{matrix}$

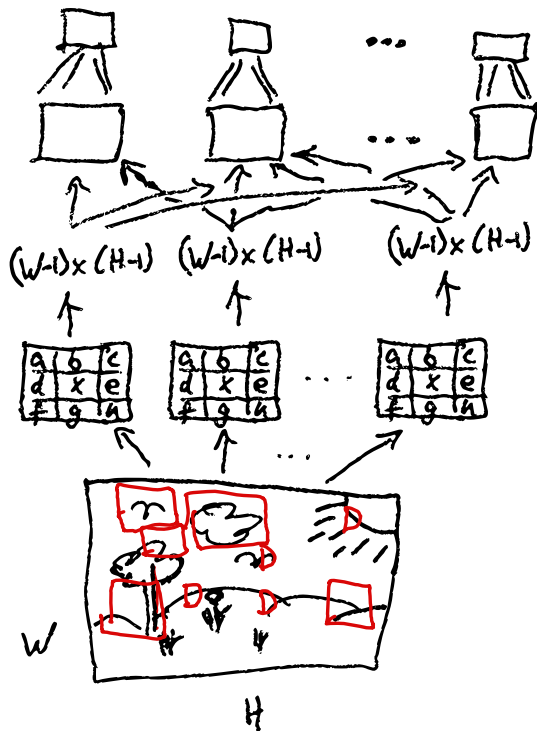
?

Pooling

L2
293x3
Conv

Pooling

L1
20
3x3
Conv
filters



Classification Problem:



Is this image taken outdoors or indoors?

Layer 1 filters.



... x20

Many possible filters were not produced:



Where do the values of the filter weights come from?

Filter 1

conv 1

a	b	c
d	x	e
f	g	h

conv k

a	b	c
d	x	e
f	g	h

$$x w_{a1} + x w_{b1} + \dots + x w_{n1}$$

+

$$x w_{a2} + x w_{b2} + \dots + x w_{n2}$$